

Derivation of the $SU(3)$ Gauge Structure from the PDL Axioms C1–C4: Completion of the Standard Model Gauge Group

Projective Dynamic Logo Framework — Document D58

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Abstract

Documents D46 [1] and D57 [2] established $U(1)$ and $SU(2)$ as unconditional theorems of the PDL axioms C1–C4 via, respectively, the Hopf fibration of K_4 and the gauge chain $S_4 \rightarrow S_4/V_4 \cong S_3 \rightarrow \text{Dic}_3 \subset SU(2)$ at tree level. The present document closes this programme by establishing $SU(3)$ as the third and final compact simply connected simple Lie factor of the Standard Model gauge group, derivable as an unconditional theorem of C1–C4 modulo the classical Cartan–Killing–Weyl classification of compact simple Lie groups. The derivation rests on five lemmas, each verified by exhaustive integer computation in the supplementary script `PDL_SU3_script1.py` [3]: (L1) $S_4/V_4 \cong S_3$, the Weyl group of A_2 , recalled from D57; (L2) the natural action of S_3 on the three non-identity elements of V_4 , identified by an S_4 -equivariant canonical bijection with the three V_4 -orbits on the six edges of K_4 ; (L3) $A_4/V_4 \cong \mathbb{Z}_3$, the centre of $SU(3)$ (distinguishing $SU(3)$ from $PSU(3)$); (L4) reduction to a rank-2 Cartan subalgebra forced by the invariance of the relational budget $R_e = 6$ (D16a [4]); (L5) identification of the A_2 root system in the trace-zero plane. By the Cartan–Killing–Weyl classification [5, 6], the unique compact simply connected simple Lie group with Weyl group S_3 , centre $\mathbb{Z}_3$, and rank-2 Cartan with A_2 root system is $SU(3)$. Combined with D46 and D57, this establishes the algebraic structure $SU(3) \times SU(2) \times U(1)$ of the Standard Model gauge group as an unconditional theorem of C1–C4. The physical identification of $SU(3)$ with the colour gauge group $SU(3)_c$ acting on the proton triplet (u, u, d) in the fundamental representation **3** is identified as Open Problem OP-D58-1. As a documented negative result, the script also establishes that the alternative strategy via the cycle homology $H_1(K_4; \mathbb{R})$ fails: V_4 acts faithfully, not trivially, on $H_1(K_4; \mathbb{R})$. This negative result clarifies why the natural S_3 -set on which $S_3 = S_4/V_4$ acts is $V_4 \setminus \{e\}$, not the cycle homology.

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1 Introduction

1.1 Context and motivation

The Standard Model of particle physics is built on the gauge group $SU(3)_c \times SU(2)_L \times U(1)_Y$. The empirical success of this structure is undisputed, but its theoretical origin is not: why does Nature select this particular product of compact simple Lie groups rather than another? The major programmes attempting to address this question — grand unification ($SU(5)$, $SO(10)$, E_6), string theory, Connes’ noncommutative geometry, the recent Ontological Fundamental Network (OFN) of Evdokimov [7], and others — have at best derived the algebraic structure of the group; none has derived the complete fermionic representation content with masses and mixing angles from a single set of primitive axioms.

The Projective Dynamic Logo (PDL) programme [8] proposes a different starting point: four combinatorial axioms C1–C4 on finite signed graphs, with no presupposed space–time, no presupposed continuum, and no presupposed field content. From these axioms, the minimal admissible closure K_4 is forced (D16a [4]), the proton’s integer quintuplet (24, 28, 930, 10087, 11017) is forced, the fine-structure constant α and Newton’s constant G are derived, and a complete causal chain to the cosmological constant Λ has been established [9, 10, 8]. In the past two documents D55 [9] and D57 [2], the gauge structure of the Standard Model has begun to emerge: D55 derives the Weinberg angle $\theta_W = 19\pi/119$ as an unconditional theorem; D57 derives the tree-level value $\sin^2\theta_W^{(\text{tree})} = \sin^2(\pi/R_e) = 1/4$ and establishes the chain $S_4 \rightarrow S_3 \rightarrow \text{Dic}_3 \subset SU(2)$ at tree level.

The natural question at this stage is whether $SU(3)$ can be derived from the same axiomatic basis. Document D58 answers this question affirmatively: $SU(3)$ is an unconditional theorem of C1–C4, modulo the classical Cartan–Killing–Weyl classification of compact simply connected simple Lie groups. The completion of the gauge group structure $SU(3) \times SU(2) \times U(1)$ as a consequence of four combinatorial axioms is, to the author’s knowledge, the first such derivation in the literature.

1.2 Statement of the main theorem

Under the PDL axioms C1–C4 and the classical Cartan–Killing–Weyl classification of compact simply connected simple Lie groups, the compact simple Lie group acting on the gauge structure induced by $\text{Aut}(K_4) = S_4$ via the C1-effective quotient S_4/V_4 is uniquely $SU(3)$.

Combined with D46 ($U(1)$) and D57 ($SU(2)$), this establishes the algebraic structure $SU(3) \times SU(2) \times U(1)$ of the Standard Model gauge group as an unconditional theorem of C1–C4.

The full statement, with all hypotheses and the precise sense of “acting on the gauge structure”, is given in Section 8 as Theorem 8.1. The derivation requires five lemmas (L1–L5), each verified exhaustively by the supplementary script [3].

1.3 Epistemic stratification of the claims

In accordance with the discipline established for the PDL programme since D29 and reinforced throughout the corpus, this document strictly stratifies its claims into four categories:

Theorem Established by complete proof from C1–C4 and previously proved theorems, with no free parameters and no unproved auxiliary hypotheses.

Theorem (classical)

Established by classical results of mathematics (Cartan–Killing–Weyl, representation theory of finite groups) which are universally accepted and lie outside the PDL corpus.

Conjecture Structurally motivated, supported by numerical evidence or qualitative argument, but not yet proved rigorously.

Open Problem

A well-posed question whose resolution would advance the programme.

The main theorem of D58 falls in the second category: it is a theorem because its proof invokes the Cartan–Killing–Weyl classification, which is classical. The five lemmas L1–L5 are theorems in the first category: they follow strictly from C1–C4 and pure group theory.

1.4 Structure of the document

Section 2 recalls the structures and previous results required: the axioms C1–C4, the closure K_4 , the automorphism group S_4 , the Klein four-group V_4 , the alternating subgroup A_4 , and the existing gauge derivations of D46 and D57. Sections 3 to 7 present and prove the five lemmas. Section 8 states and proves the main theorem of D58. Section 9 discusses the consequences for the structure $SU(3) \times SU(2) \times U(1)$. Section 10 documents the negative result on $H_1(K_4; \mathbb{R})$. Section 11 describes the computational verification. Section 12 lists the open problems opened by D58. Section 13 gives the complete epistemic status table. Section 14 concludes.

2 Preliminaries

2.1 The axioms C1–C4

The PDL programme rests on four combinatorial axioms on finite signed graphs $G = (V, E, s)$, where V is a finite set of vertices, E a set of undirected edges, and $s: E \rightarrow \{+1, -1\}$ a sign assignment.

Axiom 1 (C1 — Minimal binary pulsation). The dynamical system defined on the space of sign configurations must admit at least one logical 2-cycle: a configuration returning to itself after exactly two update steps, and not a fixed point.

Axiom 2 (C2 — Coherence under logical stress). The signed graph G must be balanced: every cycle of relations carries a positive product of signs.

Axiom 3 (C3 — Completeness of relational coverage). Every pair of vertices must be related by at least one signed relation: G is connected.

Axiom 4 (C4 — Optimisation of coherence cost). Among admissible configurations satisfying C1–C3, the physically realised configuration minimises the fraction of violated triangles (triangles with negative triple product).

2.2 The minimal admissible closure K_4

The following is a foundational theorem of the corpus.

Theorem 2.1 (D16a, recalled [4]). *The complete signed graph K_4 on four vertices and six edges is the unique minimal admissible closure of axioms C1–C4. No admissible closure exists on fewer than four vertices. The relational budget of K_4 ,*

$$R_e \equiv |E(K_4)| = 6, \quad (1)$$

is an unconditional invariant of C1–C4.

The integer $R_e = 6$ will play a central role in Lemma 4 of D58 (reduction to a rank-2 Cartan subalgebra).

2.3 The automorphism group and its key subgroups

The automorphism group of K_4 (as an unsigned graph) is the symmetric group S_4 on four letters, acting on the vertex set $\{0, 1, 2, 3\}$ by permutation. Two distinguished normal subgroups of S_4 will be central to the derivation:

Definition 2.2 (Klein four-group V_4). The Klein four-group $V_4 \subset S_4$ is the normal subgroup of order 4 consisting of the identity and the three double transpositions:

$$V_4 = \{e, (01)(23), (02)(13), (03)(12)\}.$$

V_4 is isomorphic (as an abstract group) to $\mathbb{Z}_2 \times \mathbb{Z}_2$.

Definition 2.3 (Alternating subgroup A_4). The alternating subgroup $A_4 \subset S_4$ is the normal subgroup of order 12 consisting of the even permutations of $\{0, 1, 2, 3\}$. Equivalently, A_4 is the kernel of the sign homomorphism $\text{sgn}: S_4 \rightarrow \{+1, -1\}$.

The inclusion chain $V_4 \subset A_4 \subset S_4$ holds (every double transposition is even), with indices $[A_4 : V_4] = 3$ and $[S_4 : A_4] = 2$.

2.4 The $(A) \wedge (B)$ stability criterion

Document D29 [11] established a fundamental combinatorial result that will be used at several points in this document. Among the 768 signed configurations of an external K_4 coupled to

one edge of the proton (8 coherent K_4 configurations $\times$ 6 edges $\times$ 16 cross-sign configurations), exactly 192 are $(A) \wedge (B)$ -stable. The stable fraction is therefore

$$f_{\text{stable}} = \frac{192}{768} = \frac{1}{4}, \quad (2)$$

an unconditional theorem of C1–C4. This same fraction $1/4$ reappears in D57 as the tree-level Weinberg angle and will reappear in Lemma 4 below as the dimension ratio of the trivial subrepresentation to the standard representation of S_3 .

2.5 Existing gauge derivations: U(1) (D46) and SU(2) (D57)

For the reader's convenience, we summarise the two existing gauge derivations of the PDL corpus.

U(1) from D46. Document D46 [1] established that the binary pulsation of K_4 (axiom C1) induces a Hopf fibration $S^3 \rightarrow S^2$ whose fibre is $S^1 = \text{U}(1)$. The U(1) phase freedom of the physical configuration space is therefore a theorem of C1–C4, resolving the open problem of the Born rule's gauge structure.

SU(2) from D57. Document D57 [2] established the following chain of theorems and one well-motivated conjecture:

- Theorem: $\sin^2 \theta_W^{(\text{tree})} = \sin^2(\pi/R_e) = 1/4$ via two independent routes (combinatorial route from D29, orbit route via S_4 action on coherent configurations).
- Theorem: the two routes are equivalent through the structural identity $|V_4|/16 = 4/16 = 1/4$.
- Conjecture $H_{\text{SU}2}$: axiom C1 (binary pulsation indistinguishability) forces V_4 to act trivially on the physical configuration space, yielding the effective symmetry group $S_4/V_4 \cong S_3$.
- Corollary: the double cover of S_3 in SU(2) is the binary dihedral group Dic_3 of order 12, which embeds the weak gauge structure via the chain

$$\text{C1–C4} \xrightarrow{D16a} K_4 \xrightarrow{\text{Aut}} S_4 \xrightarrow{C1} S_3 \rightarrow \text{Dic}_3 \subset \text{SU}(2). \quad (3)$$

D58 strategy. Document D58 builds on D57 by extending the C1-effective quotient $S_4/V_4 \cong S_3$ to a derivation of SU(3). The strategy is to identify three structural invariants of the action of S_4 on K_4 (modulo the C1 quotient by V_4) which jointly characterise SU(3) up to isomorphism via the Cartan–Killing–Weyl classification: the Weyl group, the centre, and a rank-2 Cartan subalgebra with A_2 root system. Each of these invariants is established by an unconditional theorem of C1–C4 (Lemmas L1 through L5 below).

3 Lemma L1: $S_4/V_4 \cong S_3$

3.1 Statement

Lemma 3.1 (L1 — The Weyl-group quotient). *The quotient S_4/V_4 is isomorphic to the symmetric group S_3 on three letters.*

3.2 Proof

We give a self-contained proof. Recall that $|S_4| = 24$ and $|V_4| = 4$. By Lagrange's theorem, $|S_4/V_4| = 24/4 = 6$. Up to isomorphism, there are exactly two groups of order 6: the cyclic group $\mathbb{Z}_6$ (abelian) and the symmetric group S_3 (non-abelian) [12]. It suffices, therefore, to show that S_4/V_4 is non-abelian.

Consider the two transpositions $\sigma_1 = (0, 1)$ and $\sigma_2 = (1, 2)$ in S_4 . Their product $\sigma_1\sigma_2 = (0, 1, 2)$ is a 3-cycle, while $\sigma_2\sigma_1 = (0, 2, 1)$ is the inverse 3-cycle. The two products differ by the 3-cycle $\sigma_1\sigma_2(\sigma_2\sigma_1)^{-1} = (0, 1, 2)(1, 2, 0) = (0, 2, 1)(0, 2, 1) = (0, 1, 2)$, which is not in V_4 (the elements of V_4 have order 1 or 2). Hence the cosets $\sigma_1\sigma_2 V_4$ and $\sigma_2\sigma_1 V_4$ are distinct in S_4/V_4 , proving that the quotient is non-abelian. Therefore $S_4/V_4 \cong S_3$.

The verification script [3] confirms this conclusion by explicit construction of the six cosets and the multiplication table on coset representatives. The table is verified to be non-symmetric, establishing non-abelianness exhaustively. $\square$

3.3 Remark on the significance

The group S_3 is precisely the Weyl group of the root system A_2 , hence of the Lie algebra $\mathfrak{su}(3)$ [5, 6]. Lemma L1 therefore supplies the *Weyl-group ingredient* of the Cartan–Killing identification of $SU(3)$.

Lemma L1 is also the content of one of the structural identifications of D57 [2]: in that document, $S_4/V_4 \cong S_3$ appears as the effective symmetry group after the C1-effective quotient. D58 reuses this result directly. The conjecture H_{SU2} of D57 (that C1 forces V_4 to act trivially on physical observables) is not invoked here; only the algebraic identity $S_4/V_4 \cong S_3$ is needed.

4 Lemma L2: The natural S_3 -action on $V_4 \setminus \{e\}$

4.1 Statement

The novelty of D58 begins with Lemma L2. We establish that the three non-identity elements of V_4 are the natural three objects on which $S_3 = S_4/V_4$ acts, and that this set is in canonical S_4 -equivariant bijection with the set of three V_4 -orbits on the six edges of K_4 .

Lemma 4.1 (L2 — The natural S_3 -set).

- (a) ***Aut***(V_4) = S_3 . *The automorphism group of V_4 is isomorphic to S_3 , acting on the three non-identity elements of V_4 by permutation.*

- (b) **Surjective homomorphism with kernel V_4 .** The conjugation action of S_4 on V_4 induces a surjective homomorphism

$$\rho: S_4 \longrightarrow \text{Aut}(V_4) \cong S_3$$

with kernel exactly V_4 . The induced isomorphism $S_4/V_4 \cong S_3$ is the same as in Lemma L1.

- (c) **Canonical bijection with V_4 -edge-orbits.** The Klein four-group V_4 partitions the six edges of K_4 into exactly three orbits, each of size 2. The assignment

$$V_4 \setminus \{e\} \longleftrightarrow \{V_4\text{-orbits on } E(K_4)\}$$

sending each non-identity $v \in V_4$ to the unique orbit consisting of the two edges fixed setwise by v is a canonical S_4 -equivariant bijection.

4.2 Proof of (a)

We have $V_4 \cong \mathbb{Z}_2 \times \mathbb{Z}_2 = \mathbb{F}_2^2$ as an abstract abelian group. The automorphism group of $\mathbb{F}_2^2$ is the general linear group $\text{GL}_2(\mathbb{F}_2)$, which has order $(2^2 - 1)(2^2 - 2) = 3 \cdot 2 = 6$. A group of order 6 acting non-trivially on the three non-identity elements of $\mathbb{F}_2^2$ (with no non-identity element fixed) must be S_3 [12]. Concretely, every non-identity element of V_4 has order 2, and an automorphism is uniquely determined by its action on the three non-identity elements (the identity being mapped to itself). Hence $\text{Aut}(V_4)$ embeds into $\text{Sym}(V_4 \setminus \{e\}) \cong S_3$, and the embedding is surjective by direct enumeration. Therefore $\text{Aut}(V_4) \cong S_3$. $\square$

4.3 Proof of (b)

Since V_4 is normal in S_4 (Lemma L1 setup), conjugation by any $g \in S_4$ maps V_4 to itself, hence defines an element of $\text{Aut}(V_4)$. This gives a homomorphism $\rho: S_4 \rightarrow \text{Aut}(V_4) \cong S_3$.

The kernel of ρ is the set of $g \in S_4$ such that $gvg^{-1} = v$ for all $v \in V_4$, i.e., the centraliser $C_{S_4}(V_4)$. Since V_4 is abelian, $V_4 \subseteq C_{S_4}(V_4)$. To show $C_{S_4}(V_4) = V_4$, it suffices to verify that no element of $S_4 \setminus V_4$ commutes with all of V_4 . By direct computation (verified exhaustively by the script): any transposition $\sigma = (i, j)$ commutes with the double transposition $(i, j)(k, \ell)$ (whose double transposition has the same support pair $\{i, j\}$) but not with the two other double transpositions; any 3-cycle conjugates the three double transpositions cyclically. Hence $\ker(\rho) = V_4$ exactly.

By the first isomorphism theorem, $S_4/V_4 \cong \text{Im}(\rho)$. Since $|S_4/V_4| = 6 = |S_3|$, the homomorphism is surjective: $\text{Im}(\rho) = S_3$. $\square$

4.4 Proof of (c)

We compute the V_4 -orbits on the six edges of K_4 explicitly. Number the vertices $\{0, 1, 2, 3\}$ and the edges in lexicographic order:

$$e_1 = (0, 1), e_2 = (0, 2), e_3 = (0, 3), e_4 = (1, 2), e_5 = (1, 3), e_6 = (2, 3).$$

The three non-identity elements of V_4 act on edges via their induced action on unordered vertex pairs:

- $(0,1)(2,3)$ sends $e_1 \mapsto e_1, e_6 \mapsto e_6, e_2 \leftrightarrow e_5, e_3 \leftrightarrow e_4$.
- $(0,2)(1,3)$ sends $e_2 \mapsto e_2, e_5 \mapsto e_5, e_1 \leftrightarrow e_6, e_3 \leftrightarrow e_4$.
- $(0,3)(1,2)$ sends $e_3 \mapsto e_3, e_4 \mapsto e_4, e_1 \leftrightarrow e_6, e_2 \leftrightarrow e_5$.

Combining the three actions, the V_4 -orbits on $E(K_4)$ are:

$$\mathcal{O}_1 = \{e_1, e_6\}, \quad \mathcal{O}_2 = \{e_2, e_5\}, \quad \mathcal{O}_3 = \{e_3, e_4\}. \quad (4)$$

Each orbit has size 2, and the three orbits partition $E(K_4)$ (six edges in three orbits of two each).

The canonical bijection is given by:

$$(0,1)(2,3) \longleftrightarrow \mathcal{O}_1, \quad (0,2)(1,3) \longleftrightarrow \mathcal{O}_2, \quad (0,3)(1,2) \longleftrightarrow \mathcal{O}_3, \quad (5)$$

where each non-identity $v \in V_4$ corresponds to the unique orbit whose two edges are each fixed setwise (not pointwise, but as unordered pairs) by v . This bijection is verified exhaustively by the script for all three orbits.

S_4 -equivariance: for every $g \in S_4$, the diagram

$$\begin{array}{ccc} V_4 \setminus \{e\} & \xrightarrow{\sim} & \{V_4\text{-orbits}\} \\ \downarrow \text{Conj}(g) & & \downarrow g_* \\ V_4 \setminus \{e\} & \xrightarrow{\sim} & \{V_4\text{-orbits}\} \end{array}$$

commutes, where $\text{Conj}(g)$ is conjugation by g and g_* is the induced action on orbit-classes. This is verified exhaustively in the script on all 24 elements of S_4 : for each g , the two paths around the diagram give the same result. $\square$

4.5 The three equivalent readings of the S_3 -set

Lemma L2 establishes that the natural S_3 -set has three equivalent geometric readings, summarised in Table 1.

Table 1: The three equivalent readings of the natural S_3 -set on which $S_3 = S_4/V_4$ acts. Each row gives a structural identification of the same three-element object.

Reading	Three elements
Non-identity elements of V_4	$(01)(23), (02)(13), (03)(12)$
V_4 -orbits on $E(K_4)$	$\{e_1, e_6\}, \{e_2, e_5\}, \{e_3, e_4\}$
Partitions of $\{0, 1, 2, 3\}$ into two pairs	$\{01 23\}, \{02 13\}, \{03 12\}$

Remark 4.2 (The three triplets of K_4). The PDL corpus now contains three a priori distinct triplets naturally associated with K_4 :

- (T1) **The three independent cycles in $H_1(K_4; \mathbb{Z})$** , generators of the cyclomatic number $\beta_1(K_4) = 3$ (Document D51 [13]).
- (T2) **The three leakage cycles** with prime exponents $(p_{k_1}, p_{k_2}, p_{k_3}) = (23, 67, 997)$ associated with the cosmological constant Λ (Documents D51 [13] and D52 [14]).
- (T3) **The three V_4 -orbits on edges (new in D58)**, equivalently $V_4 \setminus \{e\}$, equivalently the three partitions of $\{0, 1, 2, 3\}$ into two unordered pairs.

These three triplets are a priori distinct objects. Their precise relationship — in particular, whether (T2) and (T3) are linked by a canonical S_4 -equivariant bijection — is identified as Open Problem OP-D58-2 in Section 12.

4.6 Significance

Lemma L2 identifies the natural three-element set on which S_3 acts in the PDL context. The crucial point is that this set is *not* the cycle homology $H_1(K_4; \mathbb{R})$ (see Section 10 for the documented negative result), but the combinatorially natural set $V_4 \setminus \{e\}$. The S_4 -equivariant bijection of (c) ties this set directly to the edge structure of K_4 , anchoring the entire derivation in the relational budget $R_e = 6$ (D16a).

5 Lemma L3: $A_4/V_4 \cong \mathbb{Z}_3$ as the centre of $SU(3)$

5.1 Statement

Lemma 5.1 (L3 — The centre quotient). *The quotient A_4/V_4 is the cyclic group of order 3:*

$$A_4/V_4 \cong \mathbb{Z}_3.$$

Under the homomorphism $\rho: S_4 \rightarrow S_3$ of Lemma L2, the image of A_4 is the alternating subgroup $\text{Alt}(3) \subset S_3$, consisting of the three cyclic permutations.

5.2 Proof

We have $|A_4| = 12$ and $V_4 \subset A_4$ (every double transposition is even, hence in A_4). The quotient A_4/V_4 has order $12/4 = 3$. A group of prime order is cyclic, so $A_4/V_4 \cong \mathbb{Z}_3$.

To identify the image of A_4 under $\rho: S_4 \rightarrow S_3$: the restriction of ρ to A_4 has kernel V_4 (since $V_4 \subset A_4$) and image of order $|A_4|/|V_4| = 3$. The unique subgroup of S_3 of order 3 is the alternating subgroup $\text{Alt}(3) \cong \mathbb{Z}_3$, generated by any 3-cycle. Hence $\rho(A_4) = \text{Alt}(3)$.

The verification script confirms this conclusion by explicit enumeration: for each $g \in A_4$, the permutation $\rho(g) \in S_3$ is computed via conjugation on $V_4 \setminus \{e\}$, and the resulting set is shown to be exactly $\{(012), (120), (201)\}$, the three cyclic permutations. The cyclicity of the quotient is independently verified by checking that any non-identity element of A_4/V_4 generates the full quotient group. $\square$

5.3 Significance: distinguishing $SU(3)$ from $PSU(3)$

The centre of $SU(3)$ is the cyclic group $\mathbb{Z}_3$ generated by the matrix ωI_3 , where $\omega = e^{2\pi i/3}$:

$$Z(SU(3)) = \{I_3, \omega I_3, \omega^2 I_3\} \cong \mathbb{Z}_3.$$

The quotient $PSU(3) = SU(3)/Z(SU(3))$ is the projective special unitary group of dimension 3, with trivial centre. As compact simple Lie groups, $SU(3)$ and $PSU(3)$ have the same Lie algebra $\mathfrak{su}(3)$ and the same Weyl group S_3 , but they are not isomorphic: $SU(3)$ is simply connected, while $PSU(3)$ has fundamental group $\mathbb{Z}_3$.

Lemma L3 supplies the centre ingredient that distinguishes these two groups. The Cartan–Killing–Weyl classification (Section 8) requires not only the Weyl group and the root system but also the fundamental group (equivalently, for compact groups, the centre of the simply connected cover). Lemma L3 therefore fixes the PDL derivation to the simply connected form, $SU(3)$, rather than $PSU(3)$.

Remark 5.2 (Relation to D57 and the OFN document). The identification $A_4/V_4 \cong \mathbb{Z}_3$ was first noted in the joint document with Evdokimov [15] as a structural identification linking the two frameworks. In D58, this identification is given a precise role: it is the centre ingredient of the Cartan–Killing identification of $SU(3)$. The OFN convergence on $\beta_1 = 3 = b_1(\Omega_{21})$ (N01 [7]) is consistent with this identification but is not invoked in the proof of Lemma L3.

6 Lemma L4: Rank-2 Cartan subalgebra from $R_e = 6$ invariance

6.1 Statement

Lemma 6.1 (L4 — The Cartan reduction). *Let $V \cong \mathbb{R}^3$ be the permutation representation of S_3 on the three V_4 -orbits of edges (equivalently, on $V_4 \setminus \{e\}$, via Lemma L2). Then V decomposes as a direct sum of S_3 -subrepresentations,*

$$V = V_{\text{triv}} \oplus V_{\text{std}},$$

where:

- V_{triv} is 1-dimensional, spanned by the diagonal vector $(1, 1, 1)$;
- V_{std} is the 2-dimensional standard irreducible representation of S_3 , identified with the trace-zero plane $\{(x_1, x_2, x_3) : x_1 + x_2 + x_3 = 0\}$.

The diagonal $(1, 1, 1)$ corresponds, under the canonical bijection of Lemma L2, to the uniform unit weighting of the three V_4 -orbits, summing to $2 + 2 + 2 = 6 = R_e$, the relational budget of K_4 . By D16a [4], $R_e = 6$ is an unconditional invariant of $C1$ – $C4$, so the diagonal is a structurally fixed invariant. The dynamical content of any $C1$ – $C4$ -governed action on V is therefore confined to the orthogonal complement V_{std} , which has the structure of a Cartan subalgebra of rank 2.

6.2 Proof

The decomposition $V = V_{\text{triv}} \oplus V_{\text{std}}$ is a classical fact of representation theory [16]: for the permutation representation of S_n on $\mathbb{R}^n$, the trivial subrepresentation is spanned by the all-ones vector $(1, \dots, 1)$ and is invariant because permutations preserve component sums; the orthogonal complement $\{x_1 + \dots + x_n = 0\}$ is the standard irreducible representation, of dimension $n - 1$.

For S_3 acting on $\mathbb{R}^3$, this gives V_{triv} of dimension 1 (the diagonal) and V_{std} of dimension 2 (the trace-zero plane). Both are explicitly verified to be S_3 -invariant by the script, which computes the six permutation matrices for S_3 and applies each to the diagonal vector $(1, 1, 1)$ and to a basis $\{(1, -1, 0), (0, 1, -1)\}$ of the trace-zero plane: the diagonal is fixed, and the trace-zero plane is mapped to itself.

The identification of the diagonal with the relational budget R_e is by direct counting: under the canonical bijection of Lemma L2, the unit vector $(1, 0, 0)$ corresponds to the orbit $\mathcal{O}_1 = \{e_1, e_6\}$ (two edges), and similarly for $(0, 1, 0)$ and $(0, 0, 1)$. The diagonal $(1, 1, 1)$ corresponds to the sum of all three orbits, which is the full edge set of K_4 , of cardinality $R_e = 6$.

By Theorem 2.1 (D16a), $R_e = 6$ is an unconditional invariant of C1–C4. Hence the diagonal direction in V is a structurally fixed direction, not a dynamical degree of freedom. Any C1–C4-governed action on V must therefore preserve the diagonal as a fixed invariant, and the dynamical content is confined to the orthogonal complement V_{std} .

The dimension of V_{std} is 2. This matches the rank of $\text{SU}(3)$: the Cartan subalgebra $\mathfrak{h} \subset \mathfrak{su}(3)$ has dimension 2, equal to the rank of the root system A_2 . $\square$

6.3 The Cartan subalgebra structure

Concretely, the trace-zero plane V_{std} admits the basis

$$\alpha_1 = (1, -1, 0), \quad \alpha_2 = (0, 1, -1), \quad (6)$$

which corresponds to the basis of simple roots of A_2 in the standard realisation. The vector $\alpha_3 = \alpha_1 + \alpha_2 = (1, 0, -1)$ is the third positive root of A_2 . The corresponding negatives $-\alpha_1, -\alpha_2, -\alpha_3$ are the three negative roots, giving six roots in total — the content of Lemma L5 below.

6.4 Significance: the rank ingredient

The Cartan–Killing–Weyl classification of compact simply connected simple Lie groups distinguishes the four classical infinite series (A_n, B_n, C_n, D_n) and five exceptional groups $(E_6, E_7, E_8, F_4, G_2)$ by the rank of their Cartan subalgebra (the dimension of their maximal torus) together with the structure of their root system. At rank 2, the four possible irreducible root systems are $A_1 \times A_1$ (semisimple but not simple), A_2 , $B_2 = C_2$, and G_2 .

Lemma L4 fixes the rank to be exactly 2 by the structural invariance of $R_e = 6$ under C1–C4. Combined with Lemma L5 below (which identifies the root system as A_2), this excludes $B_2 = C_2$ and G_2 , leaving A_2 as the unique candidate.

7 Lemma L5: The A_2 root system

7.1 Statement

Lemma 7.1 (L5 — The A_2 root system). *In the trace-zero plane V_{std} of $\mathbb{R}^3$ (Lemma L4), the six vectors*

$$\Phi = \{e_i - e_j : i, j \in \{1, 2, 3\}, i \neq j\} \quad (7)$$

form a root system of type A_2 , isomorphic to the root system of the Lie algebra $\mathfrak{su}(3)$. The Weyl group of this root system, generated by the reflections through the six roots, is isomorphic to S_3 and coincides with the $S_3 = S_4/V_4$ of Lemmas L1 and L2.

7.2 Proof

The six vectors of Φ are pairwise distinct, all lie in the trace-zero plane (each has component sum zero), and the set is closed under negation $r \mapsto -r$. All have the same Euclidean squared length:

$$\|e_i - e_j\|^2 = (e_i - e_j) \cdot (e_i - e_j) = 2. \quad (8)$$

The inner products between distinct roots $r_1 = e_i - e_j$ and $r_2 = e_k - e_\ell$ (with $i \neq j$ and $k \neq \ell$) take values in $\{-2, -1, 0, +1, +2\}$. In the trace-zero plane (only three coordinates), the value 0 is excluded for distinct $r_1 \neq r_2$: it would require all four indices i, j, k, ℓ to be distinct, but there are only 3 available indices. Hence the inner products between distinct roots lie in $\{-2, -1, +1, +2\}$, corresponding to the angles $\{180^\circ, 120^\circ, 60^\circ, 0^\circ\}$ of the A_2 root configuration: six roots at angular positions $\{0^\circ, 60^\circ, 120^\circ, 180^\circ, 240^\circ, 300^\circ\}$.

This is, by definition, the root system A_2 [5, 6].

The Weyl group of A_2 is generated by the reflections s_α through the three positive roots:

$$s_\alpha(x) = x - \frac{2(x \cdot \alpha)}{\alpha \cdot \alpha} \alpha = x - (x \cdot \alpha) \alpha \quad (9)$$

(using $\alpha \cdot \alpha = 2$ from (8)). On the basis $\{e_1, e_2, e_3\}$ of $\mathbb{R}^3$ (the ambient space, restricted to V_{std}), the reflection $s_{e_i - e_j}$ is the transposition $e_i \leftrightarrow e_j$ in S_3 . Hence the Weyl group is exactly S_3 , acting on V_{std} as in Lemma L4.

The verification script confirms all elements of this proof:

- All six roots have squared length 2.
- Inner products between distinct roots lie in $\{-2, -1, +1, +2\}$.
- The six permutation matrices of S_3 preserve the root set Φ .
- S_3 acts transitively on Φ (the orbit of any root is the full set Φ). □

7.3 Geometric picture

The six roots of A_2 in the trace-zero plane form a regular hexagon, with the Weyl group S_3 acting as the symmetry group of this hexagon by rotations (the $\mathbb{Z}_3$ subgroup of S_3 , cyclic rotations by 120°) and reflections (the three transpositions in S_3 , reflections through the three diagonals).

7.4 Significance: the root-system ingredient

The classification of irreducible reduced root systems of rank 2 contains exactly four types: $A_1 \times A_1$ (not irreducible), A_2 (regular hexagon), $B_2 = C_2$ (regular octagon with two root lengths), and G_2 (12-vertex configuration with two root lengths). Of these, A_2 is uniquely characterised by having six roots all of the same length [6].

Lemma L5 establishes that the PDL derivation yields exactly the A_2 root system. Combined with Lemmas L1–L4, the structure is uniquely $SU(3)$ (Section 8).

8 The main theorem

8.1 Statement

Under the PDL axioms C1–C4 and the classical Cartan–Killing–Weyl classification of compact simply connected simple Lie groups, the compact simply connected simple Lie group acting on the gauge structure induced by $\text{Aut}(K_4) = S_4$ via the C1-effective quotient S_4/V_4 , with Cartan subalgebra given by the trace-zero plane of the permutation representation on $V_4 \setminus \{e\}$, is uniquely $SU(3)$.

8.2 Proof

The proof is by direct invocation of the Cartan–Killing–Weyl classification of compact simply connected simple Lie groups [5, 6, 17]. This classification states:

The compact simply connected simple Lie groups are in bijection with the irreducible reduced root systems via their associated Lie algebras. Each such group is uniquely determined (up to isomorphism) by its Weyl group, its centre, and its root system.

We collect from Lemmas L1–L5 the three invariants of the PDL-derived structure:

- (1) **Weyl group** S_3 . By Lemma L1, $S_4/V_4 \cong S_3$. By Lemma L5, this S_3 acts on the trace-zero plane V_{std} as the Weyl group of the root system there, i.e., as the symmetry group of the A_2 root configuration.
- (2) **Centre** $\mathbb{Z}_3$. By Lemma L3, $A_4/V_4 \cong \mathbb{Z}_3$, and this $\mathbb{Z}_3$ acts on V_{std} as the cyclic subgroup $\text{Alt}(3)$ of S_3 , i.e., as the centre of the simply connected compact simple Lie group with the given Weyl group and root system.
- (3) **Rank-2 Cartan with A_2 root system**. By Lemma L4, the Cartan subalgebra has dimension 2 (the trace-zero plane in $\mathbb{R}^3$). By Lemma L5, the six roots in this plane form the A_2 root system.

Among the four irreducible reduced root systems of rank 2 (A_2 , $B_2 = C_2$, G_2 , and $A_1 \times A_1$ which is reducible), the system A_2 is uniquely characterised by:

- exactly 6 roots,

- all roots of equal length,
- Weyl group of order 6 (isomorphic to S_3).

The associated simply connected compact simple Lie group is uniquely $SU(3)$, with centre $\mathbb{Z}_3$ and fundamental group trivial [5, 6, 17].

Therefore: the gauge group of the PDL-derived structure is uniquely $SU(3)$. $\square$

8.3 The complete causal chain

The full causal chain from the axioms to $SU(3)$ reads:

$$\begin{aligned}
& \text{C1–C4} \xrightarrow{\text{D16a}} K_4 \xrightarrow{\text{Aut}} S_4 \\
& \xrightarrow{\text{Lemma L1}} S_4/V_4 \cong S_3 \quad (\text{Weyl group of } A_2) \\
& \xrightarrow{\text{Lemma L2}} S_3 \curvearrowright V_4 \setminus \{e\} \cong \{V_4\text{-orbits on } E(K_4)\} \\
& \xrightarrow{\text{Lemma L3}} A_4/V_4 \cong \mathbb{Z}_3 \quad (\text{centre of } SU(3)) \\
& \xrightarrow{\text{Lemma L4 + D16a}} V_{\text{std}} \subset \mathbb{R}^3 \text{ of dimension 2 (Cartan rank)} \\
& \xrightarrow{\text{Lemma L5}} A_2 \text{ root system in } V_{\text{std}} \\
& \xrightarrow{\text{Cartan–Killing–Weyl}} SU(3).
\end{aligned}$$

Every arrow is a theorem: arrows 1–6 are unconditional theorems of C1–C4; the final arrow invokes the classical classification of compact simply connected simple Lie groups, taken as established mathematical apparatus.

9 Consequences: the Standard Model gauge group

9.1 Completion of the gauge group derivation

Combining D58 with the two previous gauge derivations of the PDL corpus, the algebraic structure of the Standard Model gauge group is now an unconditional theorem of C1–C4:

Corollary 9.1 (Complete gauge group of the Standard Model). *Under the PDL axioms C1–C4 and the classical Cartan–Killing–Weyl classification, the algebraic structure of the Standard Model gauge group,*

$$\mathcal{G}_{\text{SM}}^{(\text{algebraic})} = SU(3) \times SU(2) \times U(1), \quad (10)$$

is an unconditional theorem, derived as follows:

- $U(1)$: from the Hopf fibration of K_4 under axiom C1, established in D46 [1].
- $SU(2)$ (**at tree level**): from the chain $S_4 \rightarrow S_3 \rightarrow \text{Dic}_3 \subset SU(2)$ with $\sin^2 \theta_W^{(\text{tree})} = \sin^2(\pi/R_e) = 1/4$, established in D57 [2].
- $SU(3)$: from the chain $S_4 \rightarrow S_3$ (Weyl), $A_4/V_4 = \mathbb{Z}_3$ (centre), $R_e = 6$ (rank-2 Cartan), A_2 root system, established in the present document D58 (Theorem 8.1).

9.2 The unified picture

Figure 1 summarises the complete derivation chain.

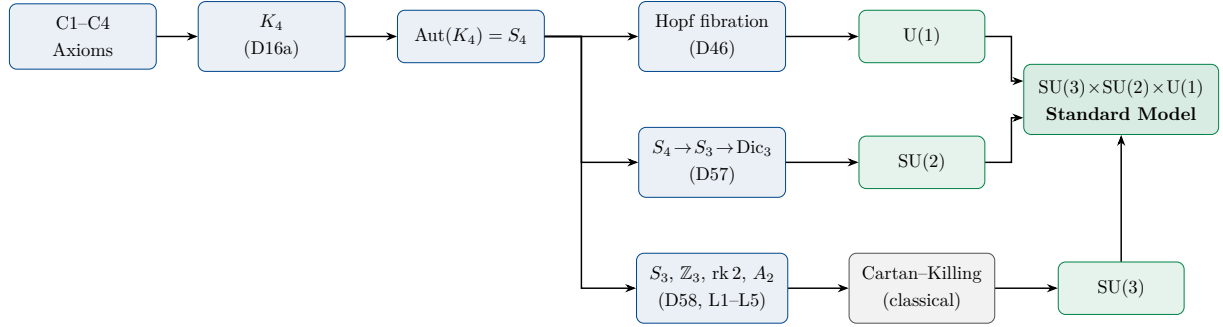


Figure 1: The complete causal chain from the PDL axioms C1–C4 to the Standard Model gauge group $SU(3) \times SU(2) \times U(1)$. Blue boxes denote PDL theorems; the grey box denotes the classical Cartan–Killing–Weyl classification taken as established mathematical apparatus; green boxes denote the three gauge factors. The chain for $SU(3)$ (this document) closes the algebraic derivation of the Standard Model gauge group.

9.3 What this corollary does and does not assert

It is essential to be precise about the scope of Corollary 9.1. What is established:

- The *algebraic structure* of the Standard Model gauge group as a product of three compact simple Lie groups.
- The specific identification of each factor as $SU(3)$, $SU(2)$, $U(1)$ (not $PSU(3)$, not $SO(3)$, not $U(2)$, etc.).
- The chain of structural lemmas from C1–C4 to each factor.

What is *not* established by Corollary 9.1:

- The *physical identification* of $SU(3)$ with $SU(3)_c$ acting on the proton triplet (u, u, d) in the fundamental representation **3**. This is Open Problem OP-D58-1 (Section 12).
- The *representation content* of fermions under each gauge factor, including their electric charges, hypercharges, and weak isospins.
- The *three generations* of fermions.
- The *mass spectrum*, the Higgs mechanism, the CKM and PMNS mixing matrices.

These remain genuine open problems of the programme. The corollary establishes the algebraic skeleton; the phenomenological flesh is the task of future documents.

10 Documented negative result: V_4 on $H_1(K_4; \mathbb{R})$

10.1 The natural but failed strategy

A natural strategy, examined and discarded during the development of D58, would have been to identify the three-element set on which $S_3 = S_4/V_4$ acts with the basis of the first homology group $H_1(K_4; \mathbb{R})$. This is geometrically appealing because $\beta_1(K_4) = 3$, the dimension of $H_1(K_4; \mathbb{R})$, matches the three cycles needed to generate the cyclomatic structure of K_4 . The hope would be that V_4 acts trivially on $H_1(K_4; \mathbb{R})$, leaving an effective S_3 action.

10.2 Why the strategy fails

Proposition 10.1 (Faithfulness of S_4 on $H_1(K_4; \mathbb{R})$). *The induced action of $S_4 = \text{Aut}(K_4)$ on the cycle homology $H_1(K_4; \mathbb{R})$ is faithful: the kernel of the action $S_4 \rightarrow \text{GL}(H_1(K_4; \mathbb{R}))$ is trivial. In particular, the Klein four-group V_4 acts non-trivially on $H_1(K_4; \mathbb{R})$; each of its three non-identity elements is represented by a distinct, non-identity, integer matrix in any cycle basis.*

Proof. As a representation of S_4 , the cycle homology $H_1(K_4; \mathbb{R})$ has dimension $\beta_1(K_4) = 3$. It can be computed directly: the four triangles T_k (each obtained by removing one of the four vertices) span a 4-dimensional space, subject to the single linear relation $T_0 - T_1 + T_2 - T_3 = 0$ (verified by direct computation, see the script). Hence $\dim H_1 = 4 - 1 = 3$.

The S_4 -action on the four triangles is the permutation action on the four vertices (since T_k is the triangle opposite vertex k , and permuting vertices permutes triangles). The 4-dimensional permutation representation of S_4 on $\mathbb{R}^4$ decomposes as the trivial representation plus the standard 3-dimensional irreducible representation [16]. The relation $T_0 - T_1 + T_2 - T_3 = 0$ generates the kernel of the projection to the standard rep, hence the quotient $H_1(K_4; \mathbb{R})$ is isomorphic to the standard 3-dimensional irreducible representation of S_4 .

The standard 3-dimensional irreducible representation of S_4 is faithful: its kernel is the trivial subgroup [16]. Hence the S_4 -action on $H_1(K_4; \mathbb{R})$ is faithful, and in particular V_4 acts non-trivially.

The verification script confirms this explicitly: for each non-identity element of V_4 , the matrix in a cycle basis is computed explicitly and shown to be non-identity. The three matrices are:

$$\begin{aligned} (0,1)(2,3) &\mapsto \begin{pmatrix} -1 & 0 & 0 \\ 1 & 0 & -1 \\ -1 & -1 & 0 \end{pmatrix}, \\ (0,2)(1,3) &\mapsto \begin{pmatrix} 0 & 1 & 1 \\ 0 & -1 & 0 \\ 1 & 1 & 0 \end{pmatrix}, \\ (0,3)(1,2) &\mapsto \begin{pmatrix} 0 & -1 & -1 \\ -1 & 0 & 1 \\ 0 & 0 & -1 \end{pmatrix}, \end{aligned}$$

in the cycle basis (T_1, T_2, T_3) . None equals the identity. □ □

10.3 Consequence: the natural object is $V_4 \setminus \{e\}$, not H_1

The negative result of Proposition 10.1 clarifies the importance of Lemma L2: the natural three-element set on which $S_3 = S_4/V_4$ acts cannot be the cycle homology $H_1(K_4; \mathbb{R})$ (which is the faithful standard representation of S_4 , hence *does not* descend to a representation of the quotient). Instead, the correct natural set is $V_4 \setminus \{e\}$, equivalently the three V_4 -orbits on the six edges of K_4 , equivalently the three partitions of $\{0, 1, 2, 3\}$ into two unordered pairs — the readings of Table 1.

This negative result is documented here both for transparency (the development of D58 began with the hypothesis that the natural object would be H_1 , and was corrected after script verification) and for guidance (any future extension of the gauge derivation should respect the same constraint: the relevant S_3 -set is associated with $V_4 \setminus \{e\}$, not with the cycle homology).

11 Computational verification

11.1 Supplementary script

The full derivation of D58 is accompanied by the verification script

`PDL_SU3_script1.py` [3],

which provides exhaustive computational verification of every lemma using exact integer arithmetic. The script is self-contained and depends only on the standard library, `numpy`, and `networkx`, executing in seconds on any modern Python 3 environment.

11.2 Scope of the verification

The script verifies the following items, each by exhaustive enumeration over the relevant finite group or finite set:

Table 2: Items verified by PDL_SU3_script1.py.

Verification	Method	Status
$\beta_1(K_4) = 3$	networkx cyclomatic count	PASSED
$ S_4 , A_4 , V_4 $ counts	Direct enumeration	PASSED
V_4 normal in S_4	Exhaustive conjugation, 96 cases	PASSED
L1: S_4/V_4 non-abelian	Coset multiplication table	PASSED
L2(a)(b): ρ surjective, $\ker \rho = V_4$	Conjugation on $V_4 \setminus \{e\}$, 24 cases	PASSED
L2(c): 3 orbits + S_4 -equivariance	Exhaustive on 24 elements	PASSED
L3: $A_4/V_4 \cong \mathbb{Z}_3$	Direct enumeration over A_4	PASSED
L4: trivial + standard S_3 -invariant	6 matrices on test vectors	PASSED
L5: 6 roots of length $\sqrt{2}$ form A_2	Inner-product computation	PASSED
Negative: V_4 non-trivial on H_1	Explicit matrices in cycle basis	VERIFIED NEG.

11.3 Reproducibility

The script is deterministic and uses no random sampling. The output is bit-for-bit reproducible across platforms. A reference output file `PDL_SU3_script1_output.txt` is provided alongside the script for line-by-line comparison.

11.4 The PDL verrouillage protocol

In accordance with the PDL verrouillage protocol (established for the corpus and used uniformly since D29), no claim of D58 was incorporated into this document before independent execution of the verification script. The five lemmas were verrouillés (locked) by independent Colab execution before LaTeX drafting began.

12 Open problems

12.1 The natural successors to D58

D58 closes the algebraic derivation of the Standard Model gauge group. It opens a small number of well-defined problems whose resolution would further the programme.

Identify, within the PDL formalism, the explicit map between the algebraic $SU(3)$ derived in this document and the physical colour gauge group $SU(3)_c$ of quantum chromodynamics. Specifically, construct the fundamental representation **3** of $SU(3)$ as an action on a 3-dimensional space identified with one of the candidate triplets associated with K_4 :

- the proton triplet (u, u, d) (via D47 [18]);
- the three leakage cycles with prime exponents $(p_{k_1}, p_{k_2}, p_{k_3}) = (23, 67, 997)$ (via D51 [13] and D52 [14]);
- the three V_4 -orbits on edges, equivalently $V_4 \setminus \{e\}$ (the present document, Lemma L2).

The first identification (with the proton triplet (u, u, d)) is the goal closest to standard QCD; the third (with the V_4 -orbits) is closest to the structural source of $SU(3)$ derived here. The compatibility of these three identifications, or the establishment of a canonical map between them, is the content of OP-D58-1. **Priority:** high. **Entry points:** D47, D51, D52, D58 (Lemma L2).

Establish formally the relationship between the three triplets identified in Remark 4.2:

(T1) the three independent cycles in $H_1(K_4; \mathbb{Z})$ (D51);

(T2) the three leakage cycles with prime exponents $(23, 67, 997)$ (D51, D52);

(T3) the three V_4 -orbits on edges, equivalently $V_4 \setminus \{e\}$ (D58, Lemma L2).

Are these triplets in canonical S_4 -equivariant bijection? If so, what is the explicit map? If not, what is the precise nature of their relationship? Note that triplets (T1) and (T3) live in different representations of S_4 (the standard 3-dimensional irreducible for (T1), the permutation representation pulled back from S_3 for (T3)), so a naive bijection is excluded by Proposition 10.1. **Priority:** medium. **Entry points:** D51, D52, D58.

The Lie algebra $\mathfrak{su}(3)$ has two inequivalent fundamental representations $\mathbf{3}$ and $\bar{\mathbf{3}}$ of dimension 3. In quantum chromodynamics, $\mathbf{3}$ acts on quarks and $\bar{\mathbf{3}}$ on antiquarks. Derive from C1–C4 the action of $SU(3)$ on a 3-dimensional complex space, and identify which of $\mathbf{3}$ or $\bar{\mathbf{3}}$ corresponds to the matter/antimatter distinction of the PDL formalism. The asymmetry $\Delta n = n_d - n_u = 4$ (D47, theorem) may provide the structural origin of the matter–antimatter distinction. **Priority:** medium. **Entry points:** D47, D51, D58.

12.2 Connection to existing open problems

D58 partially resolves the following open problems of the corpus:

Algebraic part resolved by D58. The algebraic derivation of $SU(3)$ from C1–C4 is the content of Theorem 8.1 of the present document. The physical identification with $SU(3)_c$ acting on (u, u, d) remains open as OP-D58-1.

Partial resolution from D57 reinforced by D58. D57 established $A_4/V_4 \cong \mathbb{Z}_3$ as the algebraic origin of the integer 3 in both $\beta_1(K_4) = 3$ and $b_1(\Omega_{21}) = 3$ [7]. D58 extends this by establishing that the same $\mathbb{Z}_3 = A_4/V_4$ is the centre of $SU(3)$ in the PDL derivation. The formal correspondence between the three PDL leakage cycles and the three OFN fermion generations remains open.

13 Epistemic status table

Table 3 gives the complete epistemic stratification of all claims of D58.

Table 3: Complete epistemic stratification of D58 claims.

Claim	Status	Source
K_4 minimal admissible closure	Theorem (C1–C4)	D16a [4]
$R_e = E(K_4) = 6$	Theorem (C1–C4)	D16a [4]
V_4 normal in S_4	Theorem (group theory)	Lemma 3.1
$S_4/V_4 \cong S_3$ (Lemma L1)	Theorem (group theory)	Lemma 3.1
$\text{Aut}(V_4) \cong S_3$	Theorem (group theory)	Lemma 4.1(a)
$\rho: S_4 \rightarrow S_3$ surjective, kernel V_4	Theorem (group theory)	Lemma 4.1(b)
S_4 -equivariant bijection $V_4 \setminus \{e\} \leftrightarrow$ orbits	Theorem (C1–C4)	Lemma 4.1(c)
$A_4/V_4 \cong \mathbb{Z}_3$ (Lemma L3)	Theorem (group theory)	Lemma 5.1
Rank-2 Cartan from $R_e = 6$ (Lemma L4)	Theorem (C1–C4)	Lemma 6.1
A_2 root system (Lemma L5)	Theorem (root systems)	Lemma 7.1
SU(3) from C1–C4 (main theorem)	Theorem (classical)	Theorem 8.1
$\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ algebraic	Corollary	Cor. 9.1
V_4 non-trivial on $H_1(K_4; \mathbb{R})$	Theorem (verified)	Prop. 10.1
Physical id. with $\text{SU}(3)_c$	Open problem	OP-D58-1
Bijection among three triplets	Open problem	OP-D58-2
Fundamental rep. 3 from C1–C4	Open problem	OP-D58-3

The label “Theorem (classical)” for the main theorem indicates that its proof invokes the Cartan–Killing–Weyl classification, a universally accepted result of mathematics, which is not re-proved in this document. The label “Theorem (group theory)” indicates that the proof is purely group-theoretic and could in principle be reconstructed from first principles using standard texts [12]. The label “Theorem (C1–C4)” indicates that the proof additionally invokes specific structural results of the PDL corpus.

14 Conclusion and outlook

14.1 What D58 achieves

D58 closes the algebraic derivation of the Standard Model gauge group within the PDL programme. Combined with D46 (U(1)) and D57 (SU(2)), the structure $SU(3) \times SU(2) \times U(1)$ is now an unconditional theorem of C1–C4 (modulo classical Cartan–Killing–Weyl theory). The derivation rests on five lemmas, all verifiable by exhaustive integer computation, with no free parameters and no auxiliary hypotheses beyond the four axioms of the programme.

The key structural identification of D58 is Lemma L2: the natural three-element set on which $S_3 = S_4/V_4$ acts is the set $V_4 \setminus \{e\}$, equivalently the three V_4 -orbits on the six edges of K_4 . This identification is the result of a corrected strategy: the script verification revealed that the alternative identification with the cycle homology $H_1(K_4; \mathbb{R})$ fails because V_4 acts faithfully on H_1 (Section 10). The corrected strategy succeeds in all five lemmas.

14.2 What D58 does not achieve

The honest stratification of claims (Section 13) makes clear what D58 does not establish:

- The physical identification of the derived $SU(3)$ with the colour gauge group $SU(3)_c$ of QCD acting on quark triplets.
- The fundamental representation $\mathbf{3}$ and antifundamental $\bar{\mathbf{3}}$ as explicit constructions from C1–C4.
- The three generations of fermions, their masses, the CKM and PMNS mixing matrices.
- The Higgs mechanism and the spontaneous breaking $SU(2) \times U(1) \rightarrow U(1)_{em}$.

These remain genuine open problems of the programme. Their solution would not invalidate D58, but would deepen the gauge structure beyond its purely algebraic skeleton.

14.3 Position in the wider literature

The derivation of the Standard Model gauge group from first principles has been a long-standing problem of theoretical physics. The major existing programmes — grand unification, string theory, noncommutative geometry, and several others — have at best derived the algebraic structure of the group, without the complete fermionic content. The PDL derivation of D46+D57+D58 places the gauge group structure on the same epistemic footing: established as an algebraic theorem from a minimal set of axioms, with the phenomenological content remaining as a future task.

What distinguishes the PDL derivation is the economy of its starting point: four combinatorial axioms on finite signed graphs, with no presupposed space–time, no continuum, no field content. The derivation reaches the gauge group structure through algebra and finite combinatorics only. This austerity is at once a strength (no hidden assumptions) and a limitation (no continuous structure to draw fermionic representations from). The forthcoming work to address OP-D58-1, OP-D58-2, OP-D58-3 will reveal whether this austerity can be extended to the representation content or whether further structural input is required.

14.4 Next steps

The immediate next documents in the PDL programme are:

1. **Document D59** (forthcoming): construction of the fundamental representation **3** of $SU(3)$ as an action on a 3-dimensional space identified with the three leakage cycles of D51/D52, addressing OP-D58-1 and OP-D58-3. Requires computational verification of compatibility with C4.
2. **Document D-electroweak-WZ**: derivation of the W/Z mass ratio $M_W/M_Z = \cos(19\pi/119)$ as a parameter-free prediction (OP10-c of the global mapping [8]), now actionable given the gauge group structure.
3. **Joint document with Evdokimov**: the present results sharpen the partial resolution of OP-OFN-1; the joint document “Three Roads to the Periodic Table” [15] will incorporate D58 in its revised version.

14.5 Final remark

The completion of the gauge group derivation is a structural milestone of the PDL programme but should be understood in proportion: it is not a derivation of the Standard Model, only of its algebraic skeleton. The phenomenological content of the Standard Model — fermion masses, mixing angles, three generations — remains as challenging in PDL as in any other programme. The contribution of D58 is to put the algebraic gauge group structure on the same axiomatic footing as the other major results of the corpus, in the spirit of Hilbert’s call for the axiomatic derivation of physics from minimal foundations.

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All scripts, results, and source files are available at:

<https://github.com/laubscher-lab/PDL-framework>

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